written narrative

-

Data Tidying/Wrangling

```r
# Loading needed libraries
#| warning: false
library(tidyverse)
```

```
Warning: package 'tidyverse' was built under R version 4.3.3

Warning: package 'ggplot2' was built under R version 4.3.3

Warning: package 'tidyr' was built under R version 4.3.3

Warning: package 'readr' was built under R version 4.3.3

Warning: package 'purrr' was built under R version 4.3.3

Warning: package 'dplyr' was built under R version 4.3.3

Warning: package 'stringr' was built under R version 4.3.3

Warning: package 'lubridate' was built under R version 4.3.3

-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.1.4     v readr     2.1.5
v forcats   1.0.0     v stringr   1.5.1
v ggplot2   3.5.1     v tibble    3.2.1
v lubridate 1.9.3     v tidyr     1.3.1
v purrr     1.0.2
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to
```

```r
library(readxl)
```

```
Warning: package 'readxl' was built under R version 4.3.3
```

```r
library(dplyr)
```

- **Read data from excel spreadsheet**

```r
data <- read_excel("OP-18_Waste_Minimization_and_Diversion_2020.xlsx")
```

- **Converting 'chr' to 'numeric'**

```r
df <- data

df$'Materials recycled, performance year' <- as.numeric(df$'Materials recycled, performa
df$'Materials recycled, baseline year' <- as.numeric(df$'Materials recycled, baseline ye
df$'Materials composted, performance year' <- as.numeric(df$'Materials composted, perfor
df$'Materials composted, baseline year' <- as.numeric(df$'Materials composted, baseline
df$'Materials donated or re-sold, performance year' <- as.numeric(df$'Materials donated
df$'Materials donated or re-sold, baseline year' <- as.numeric(df$'Materials donated or
df$'Materials disposed through post-recycling residual conversion, performance year' <-
df$'Materials disposed through post-recycling residual conversion, baseline year' <- as.
df$'Materials disposed in a solid waste landfill or incinerator, performance year' <- as
df$'Materials disposed in a solid waste landfill or incinerator, baseline year' <- as.nu
df$'Total waste generated, performance year' <- as.numeric(df$'Total waste generated, pe
df$'Total waste generated, baseline year' <- as.numeric(df$'Total waste generated, basel
df$'Start date, performance year or 3-year period' <- as.numeric(df$'Start date, perform
df$'End date, performance year or 3-year period' <- as.numeric(df$'End date, performance
df$'Start date, baseline year or 3-year period' <- as.numeric(df$'Start date, baseline y
df$'End date, baseline year or 3-year period' <- as.numeric(df$'End date, baseline year
df$'Number of students resident on-site, performance year' <- as.numeric(df$'Number of s
df$'Number of students resident on-site, baseline year' <- as.numeric(df$'Number of stud
df$'Number of employees resident on-site, performance year' <- as.numeric(df$'Number of
df$'Number of employees resident on-site, baseline year' <- as.numeric(df$'Number of emp
df$'Number of other individuals resident on-site, performance year' <- as.numeric(df$'Nu
df$'Number of other individuals resident on-site, baseline year' <- as.numeric(df$'Numbe
df$'Total full-time equivalent student enrollment, performance year' <- as.numeric(df$'T
df$'Total full-time equivalent student enrollment, baseline year' <- as.numeric(df$'Tota
df$'Full-time equivalent of employees, performance year' <- as.numeric(df$'Full-time equ
df$'Full-time equivalent of employees, baseline year' <- as.numeric(df$'Full-time equiva
df$'Full-time equivalent of students enrolled in exclusively in distance education, perf
df$'Full-time equivalent of students enrolled exclusively in distance education, baselin
df$'Weighted campus users, performance year' <- as.numeric(df$'Weighted campus users, pe
```

```
df$'Weighted campus users, baseline year' <- as.numeric(df$'Weighted campus users, basel
df$'Total waste generated per weighted campus user, performance year' <- as.numeric(df$'
df$'Total waste generated per weighted campus user, baseline year' <- as.numeric(df$'Tot
df$'Weighted campus users, performance year' <- as.numeric(df$'Weighted campus users, pe
df$'Weighted campus users, baseline year' <- as.numeric(df$'Weighted campus users, basel
df$'Percentage reduction in total waste generated per weighted campus user from baseline
df$'Percentage of materials diverted from the landfill or incinerator by recycling, comp
df$'Percentage of materials diverted from the landfill or incinerator (including up to 1
```

- **Converting 'chr' to 'date'**

```
df$'Last Updated' <- as.Date(df$'Last Updated')
df$'Start date, performance year or 3-year period' <- as.Date(df$'Start date, performanc
df$'End date, performance year or 3-year period' <- as.Date(df$'End date, performance ye
df$'Start date, baseline year or 3-year period' <- as.Date(df$'Start date, baseline year
df$'End date, baseline year or 3-year period' <- as.Date(df$'End date, baseline year or
```

- **Converting df from wide format to long format**

```
df_long <- df |>
  select(Institution, where(is.numeric)) |>
  pivot_longer(
    cols = starts_with("Materials") | starts_with("Total") | starts_with("Number") | sta
    names_to = c("Variable_Type", "Year"),
    names_sep = ", ",
    values_to = "Amount"
  )
```

- **Combining Tidy Numeric Data with Non Numeric Data**

```
df_non_numeric <- df |>
  select(where(~ !is.numeric(.)))

data_test <- left_join(df_long, df_non_numeric, by = "Institution")
```

- **Testing filtering by a university in institution**

```
filter_test <- data_test |>
  filter(Institution == "American University")
```

Summary Statistics

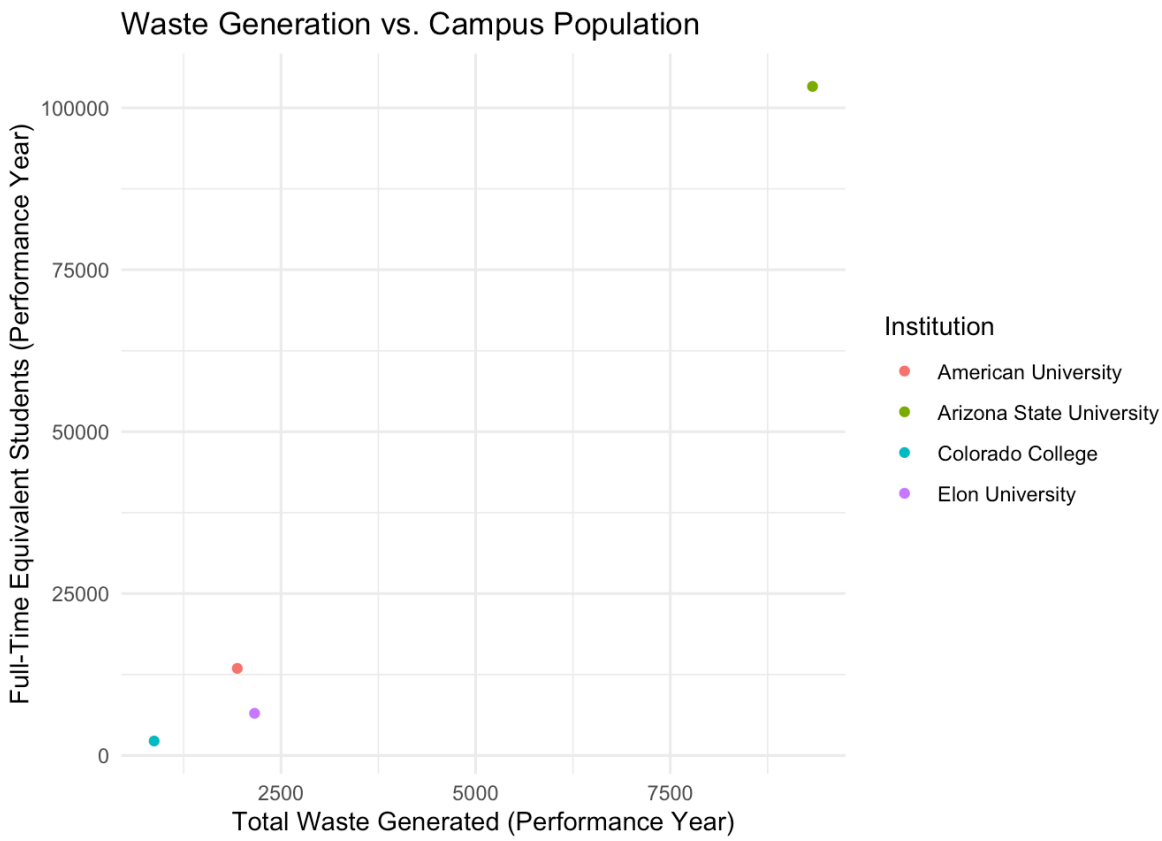
Summary

```
peer_schools<-c("Bemidji State University","Carleton College","College of St. Benedict/St. Jo
peers <-  df|>
  filter(Institution %in% peer_schools)
# Summary statistics for all numeric variables
numeric_summary <- peers |>
  summarize(across(where(is.numeric), list(
    mean = ~ mean(.x, na.rm = TRUE),
    median = ~ median(.x, na.rm = TRUE),
    sd = ~ sd(.x, na.rm = TRUE),
    min = ~ min(.x, na.rm = TRUE),
    max = ~ max(.x, na.rm = TRUE)
  )))
View(peers)
View(numeric_summary)
```

As we were comparing the University of St. Thomas with all the summary statistics of other institutions, we realized that the University of Minnesota Twin Cities significantly impacts the results. Since the University of Minnesota Twin Cities is much larger compared to other institutions, it skews the data, causing St. Thomas to appear below the average in several statistics. The main reason for this difference is likely caused by the fact that the University of Minnesota Twin Cities is much larger compared to other institutions, which naturally leads to higher numbers. St. Thomas's rank tends to fluncuate between 2nd and 3rd place in comparison to the University of Minnesota Twin Cities; it often competes with Carleton College, which has similar statistics across most variables.

Data Visuilzations

Waste Generation vs. Campus Population



This graph compares waste generations with student populations across institutions based on the performance year. The x-axis represent the total waste generated, while the y-axis show the student populations, categorized by the full-time student status. The graph is trying to show the relationship between waste generation and student population, highlighting patterns among American University, Arizona State University, Colorado College, and Elon University.

Code

```
#| warning: false
  # Filter waste and enrollment data separately, then join them
  waste_data <- data_test |>
    filter(Institution %in% c("American University", "Arizona State University", "Colorado
```

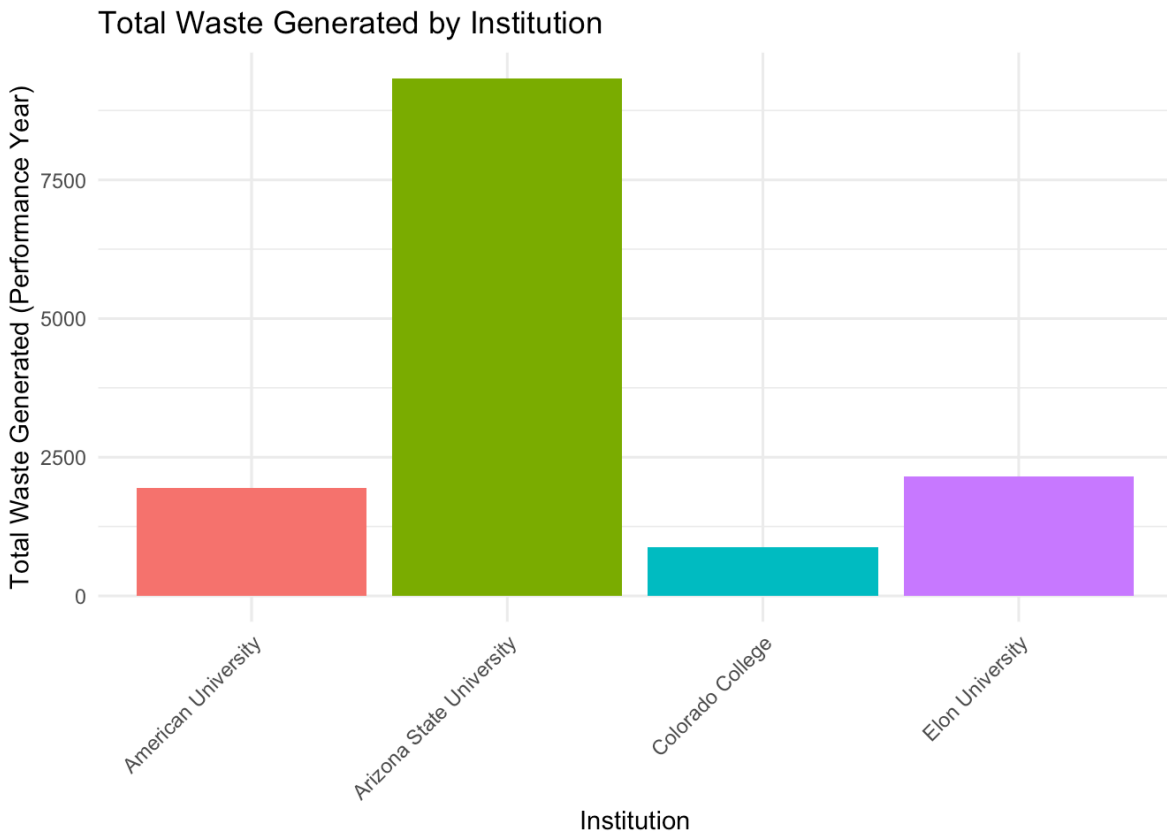
```
  filter(Variable_Type == "Total waste generated") |>
  filter(Year == "performance year") |>
  filter(!is.na(Amount)) |>
  select(Institution, Amount, Variable_Type, Year)

enrollment_data <- data_test |>
  filter(Institution %in% c("American University", "Arizona State University", "Colorado
  filter(Variable_Type == "Total full-time equivalent student enrollment") |>
  filter(Year == "performance year") |>
  filter(!is.na(Amount)) |>
  select(Institution, Amount, Variable_Type, Year)

# Join the two datasets on Institution
merged_data <- left_join(waste_data, enrollment_data, by = "Institution", suffix = c("_wa

# Plot the data
ggplot(merged_data, aes(x = Amount_waste, y = Amount_enrollment)) +
  geom_point(aes(color = Institution)) +
  labs(title = "Waste Generation vs. Campus Population",
       x = "Total Waste Generated (Performance Year)",
       y = "Full-Time Equivalent Students (Performance Year)") +
  theme_minimal()
```

Total Waste Generated by Institution



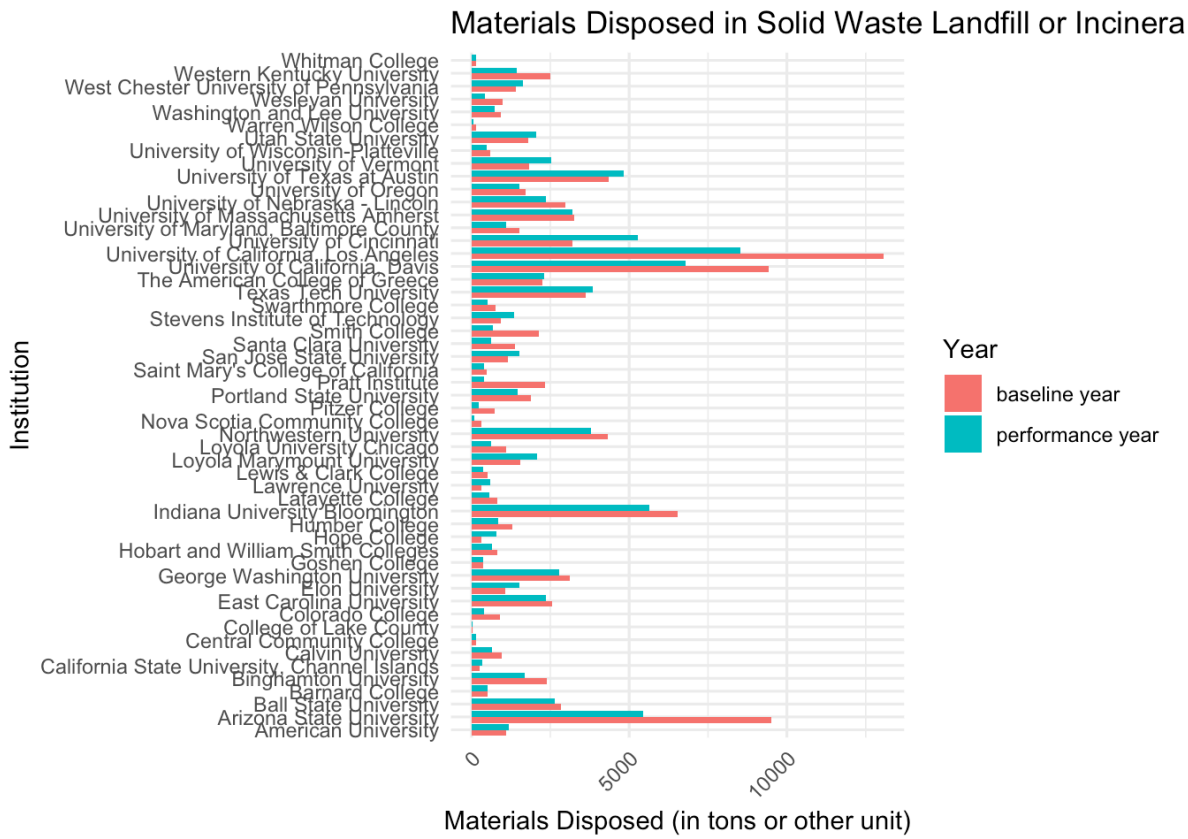
This graph is comparing total waste generated across American University, Arizona State University, Colorado College, and Elon University. The x-axis represents the institutions, while the y-axis shows the total waste generated by each institution, based on the performance year. As you can see in the bar plot, Arizona State University generates significantly more waste compared to the other three institutions."

Code

```
# Bar Plot: Total Waste Generated by Institution
ggplot(merged_data, aes(x = Institution, y = Amount_waste, fill = Institution)) +
  geom_bar(stat = "identity", show.legend = FALSE) +
  labs(title = "Total Waste Generated by Institution",
       x = "Institution",
       y = "Total Waste Generated (Performance Year)") +
```

```
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

Materials Disposed in Solid Waste Landfill or Incinerator

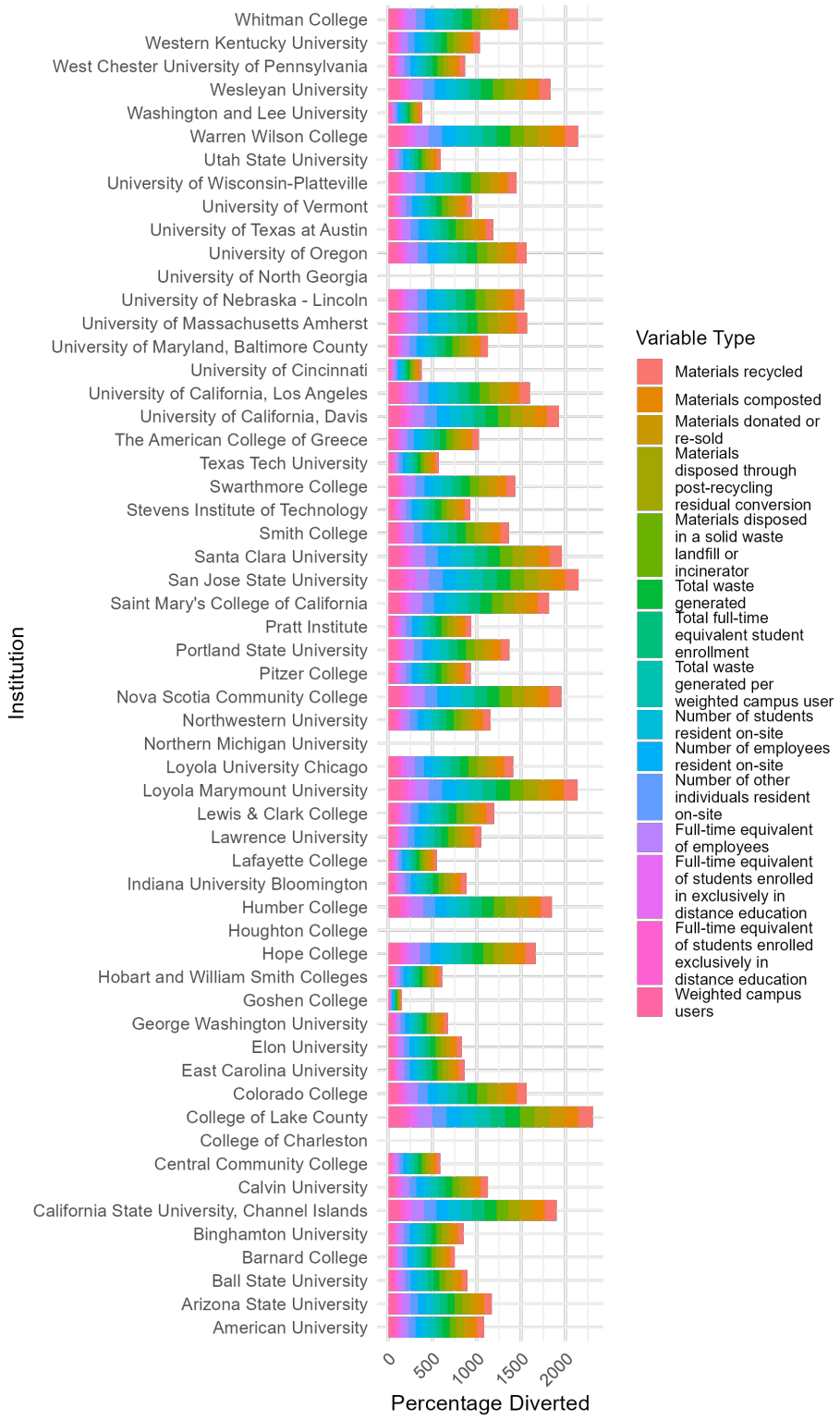


This data visualization contains two bar plots: one for the baseline year and one for the performance year, representing the disposal of materials in solid waste landfill or incinerator. The x-axis for both bar plots represent the institutions, while the y-axis show the amount of materials disposed of in tons(or other units of measurements). As seen in the graph, many of the longer bars in reference to higher disposal amounts are appearing within the baseline year. From this we can infer that higher volume of materials are likely being disposed of compared to the performance year among institutions.

Code

```
# Ensure data includes the baseline year and relevant materials information
ggplot(data_test |>
         filter(Variable_Type == "Materials disposed in a solid waste landfill or incinerato
         filter(Year %in% c("performance year", "baseline year")) |>
         filter(!is.na(Amount)),
       aes(x = Institution, y = Amount, fill = Year)) +
  geom_bar(stat = "identity", position = "dodge", show.legend = TRUE) +
  labs(title = "Materials Disposed in Solid Waste Landfill or Incinerator (Performance vs. Ba
       x = "Institution",
       y = "Materials Disposed (in tons or other unit)") +
  theme_minimal() +
  coord_flip()+
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

Institutional Waste Diversion Performance by Material Type



The graph is visualizing institutional waste diversion performance by material type. The x-axis represent different institutions, while the y-axis shows the percentage of materials diverted from landfill or incinerator, categorized by material type. In this visualization you can see that each material type is being represented by different color, providing an easier comparison of waste diversion performance across institutions.

Code

```
library(scales)
library(stringr)
# Convert 'Institution' and 'Variable_Type' to factors to ensure they work well with geom_ti
data_test$Institution <- as.factor(data_test$Institution)
data_test$Variable_Type <- as.factor(data_test$Variable_Type)
# Wrap the text in Variable_Type to create line breaks for legend labels
data_test$Variable_Type <- factor(
  str_wrap(data_test$Variable_Type, width = 30),
  levels = unique(str_wrap(data_test$Variable_Type, width = 30))
)
wrapped_title <- str_wrap("Institutional Waste Diversion Performance by Material Type", widt
# Plot using geom_col to visualize the categories
ggplot(data_test, aes(x = Institution, y = `Percentage of materials diverted from the landfi
  geom_col() +
  theme_minimal() +
  labs(
    title = wrapped_title,
    x = "Institution",
    y = "Percentage Diverted",
    fill = "Variable Type",
    caption = "Source: AASHE STAR"
  ) +
  theme(
    legend.text = element_text(size = 8),  # Make legend text smaller
    legend.title = element_text(size = 10), # Adjust legend title size
    legend.key.size = unit(0.5, "cm"),  # Reduce legend key size
    legend.spacing.y = unit(0.2, "cm")    # Reduce space between legend items
  ) +
  coord_flip()
```